

Knowledge, Logic, and Reasoning

Propositional Logic, Inference, and Satisfiability

KB

Knowledge base — facts and rules about the world

SAT

Satisfiability — is there a model that makes it all true?

From Search to Explicit Knowledge

- So far: search finds a path or assignment, guided by heuristics
- A CSP already looks like logic — variables, domains, constraints
- What if we could state facts and rules directly, and let inference do the work?
- **That's the shift from search to knowledge-based reasoning**

Knowledge-Based Agents

Declarative approach

Tell the agent what it needs to know

The agent reasons to decide what to do

Knowledge base = domain-specific facts

Inference engine = general reasoning

Inference vs. search

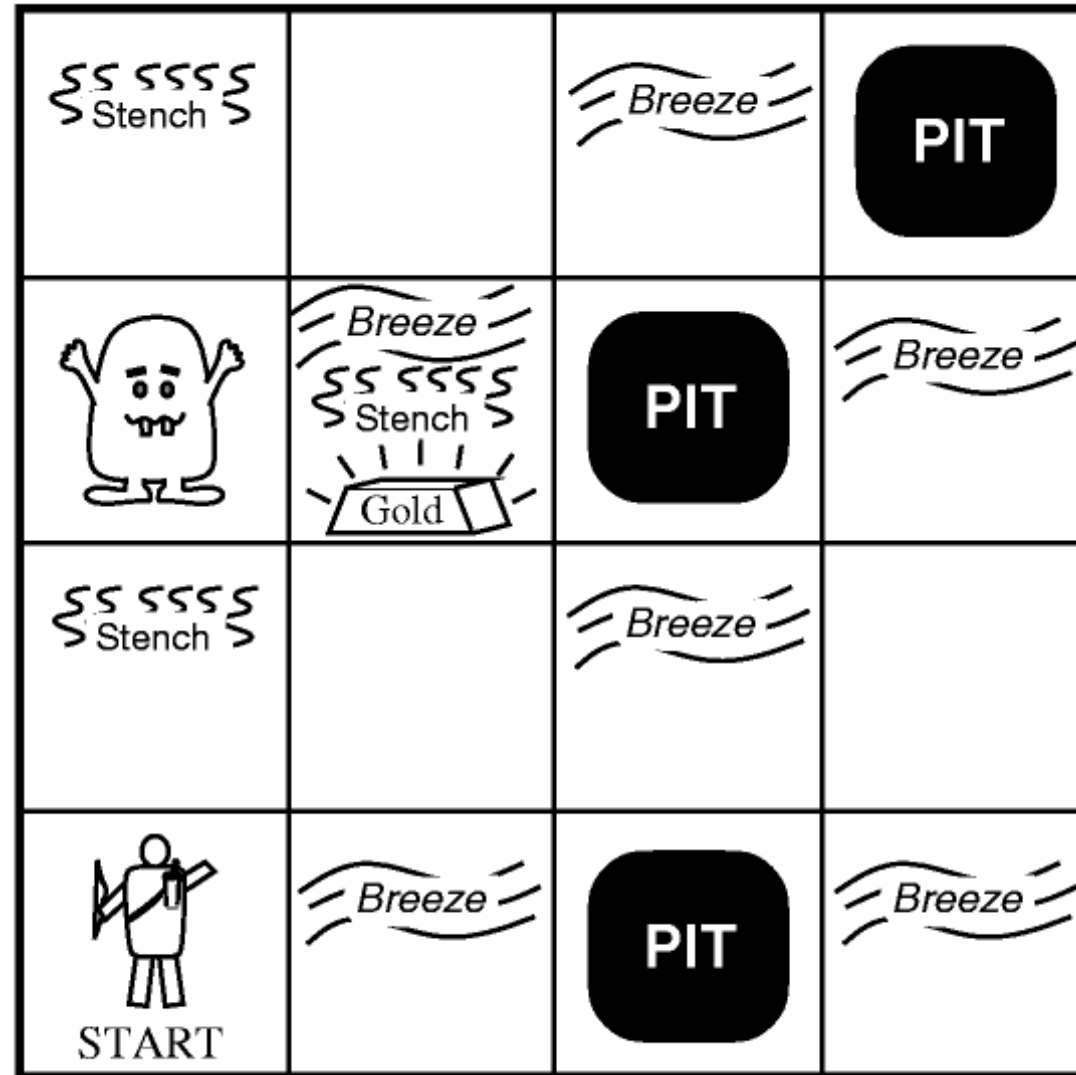
Inference: what follows from the KB?

Search: how do I get from A to B?

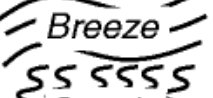
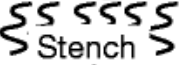
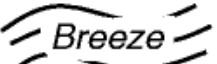
A knowledge base is a set of sentences
in a formal language

A Wumpus World

- The agent starts at (1,1) and must find the gold while avoiding pits and the wumpus
- Stench in a square → the wumpus is in an adjacent square
- Breeze in a square → a pit is in an adjacent square
- **The agent only perceives its current square — everything else must be inferred**



A Wumpus World

 Stench		 Breeze	 PIT
	 Breeze  Stench  Gold	 PIT	 Breeze
 Stench		 Breeze	
 START	 Breeze	 PIT	 Breeze

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1  OK	2,1 OK	3,1	4,1

(a)

-  = Agent
- B = Breeze
- G = Glitter, Gold
- OK = Safe square
- P = Pit
- S = Stench
- V = Visited
- W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1  B OK	3,1 P?	4,1

(b)

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2  S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

-  = Agent
- B = Breeze
- G = Glitter, Gold
- OK = Safe square
- P = Pit
- S = Stench
- V = Visited
- W = Wumpus

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3  S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

A Worked Example: Modus Ponens in Action

- **Percepts collected so far:**

- $\neg S(1,1), \neg S(2,1), S(1,2)$ — no stench at (1,1)/(2,1), stench at (1,2)
- $\neg B(1,1), B(2,1), \neg B(1,2)$ — no breeze at (1,1)/(1,2), breeze at (2,1)

- **Rule r1 (wumpus causes stench):**

- r1: $\neg S(1,1) \Rightarrow \neg W(1,1) \wedge \neg W(1,2) \wedge \neg W(2,1)$

Modus Ponens with $\neg S(1,1)$ and r1 gives:

$\neg W(1,1) \wedge \neg W(1,2) \wedge \neg W(2,1)$

And-Elimination gives: $\neg W(1,1), \neg W(1,2), \neg W(2,1)$

PROPOSITIONAL-KB-AGENT(percept)

static: KB, a knowledge base

t, a counter, initially 0

TELL(KB, MAKE-PERCEPT-SENTENCE(percept,t))

for each action in possible actions do

if ASK(KB, MAKE-ACTION-QUERY(t,action))

then $t \leftarrow t+1$; return action

Syntax, Semantics, and Truth

- Syntax: defines the form of sentences — which expressions are well-formed
- Semantics: defines the meaning — in which worlds a sentence is true
- A possible world is a complete assignment of truth values to all symbols
- **Every sentence is TRUE or FALSE in a given world — never both**

Logic of Atomic Propositions

Definition 1.7 The alphabet of propositional logic consists of letters $p, q, \dots$, denoting elementary statements. And also special symbols $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$, and brackets $(,)$.

A word in this alphabet is called a formula if it satisfies the following conditions:

1. an elementary statement is a formula;
2. if A and B are formulas, then $(\neg A)$, $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$ are also formulas;³
3. only words satisfying conditions 1 and 2 are formulas.

³Note that the symbols A and B are not part of the alphabet. They are symbols of the metalanguage; for example, A can replace the formula $(p \wedge q)$.

1

Atomic propositions

P: "It is raining", Q: "The ground is wet"

2

Negation, conjunction, disjunction

$\neg P$, $P \wedge Q$, $P \vee Q$

p	$\neg p$
T	F
F	T

p	$\neg p$
1	0
0	1

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \vee q$
1	1	1
1	0	1
0	1	1
0	0	0

3

Implication, biconditional (equivalence)

$P \Rightarrow Q$, $P \Leftrightarrow Q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

p	q	$p \leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

Table 1.6: Priority of Logical Operations

Operator	Order
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Logical Equivalences Reference

Definition 1.8 *A complex statement A is called a tautology if it is true for any values of the input variables.*

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T	T	T
T	F	F	T	T	T	T	F
F	T	T	F	F	F	F	T
F	F	T	T	T	T	T	T

Basic Equivalences	
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Laws of identity
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Laws of domination
$p \vee p \equiv p$ $p \wedge p \equiv p$	Laws of idempotence
$\neg(\neg p) \equiv p$	Law of double negation
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Laws of negation

Logical Equivalences Reference

Equivalences with $\Rightarrow$

$$p \Rightarrow q \equiv \neg p \vee q$$

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p \quad (\text{contrapositive})$$

$$\neg(p \Rightarrow q) \equiv p \wedge \neg q$$

$$(p \Rightarrow q) \wedge (p \Rightarrow r) \equiv p \Rightarrow (q \wedge r)$$

$$(p \Rightarrow r) \wedge (q \Rightarrow r) \equiv (p \vee q) \Rightarrow r$$

Equivalences with $\Leftrightarrow$

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

$$p \Leftrightarrow q \equiv \neg p \Leftrightarrow \neg q$$

$$p \Leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \Leftrightarrow q) \equiv p \Leftrightarrow \neg q$$

A statement true for every input is a tautology.

From Propositions to Predicates

- A predicate is a statement containing a variable; the values for which it is true or false form its domain
- Example: $M(x)$ = “x is mortal”, so $M(\text{Socrates})$ = “Socrates is mortal”
- $\forall x, \exists x$ let us quantify over an entire domain instead of naming objects one at a time

Example: “All men are mortal”

$(\forall x)(P(x) \rightarrow M(x))$, where $P(x)$ = “x is a human”, $M(x)$ = “x is mortal”

Writing $(\forall x)P(x) \wedge M(x)$ instead would be wrong: true only if every object in the domain were human — the implication is what restricts “mortal” to apply just to the humans.

Terms, Formulas, and Scope

- Terms: individual variables x_i and individual constants a_i are terms
- Atomic formula: if $P^{(n)}$ is an n -ary predicate and $t_1, \dots, t_n$ are terms, then $P^{(n)}(t_1, \dots, t_n)$ is a formula
- If A, B are formulas, then $\neg A$ and $(A \rightarrow B)$ are formulas; if A is a formula and x a variable, then $(\forall x)A$ is a formula
- **A variable occurrence is bound if it lies in the scope of a quantifier $(\forall x)$; otherwise free — a formula with no free variables is closed (a sentence)**
- A term t is substitutable for x in $A(x)$ if substituting it does not accidentally capture a variable of t under a quantifier

Axiom Schemas for First-Order Logic

Propositional schemas

1. $A \rightarrow (B \rightarrow A)$
2. $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$
3. $(\neg B \rightarrow \neg A) \rightarrow ((\neg B \rightarrow A) \rightarrow A)$

Quantifier schemas

4. $(\forall x)A(x) \rightarrow A(t)$, t substitutable
for x in $A(x)$ (instantiation)
5. $(\forall x)(B \rightarrow A(x)) \rightarrow (B \rightarrow (\forall x)A(x))$,
 x not free in B (quantifier intro)

Rules of Inference and Derivability

Modus ponens:

$$\frac{\Gamma \vdash A, \quad \Gamma \vdash A \rightarrow B}{\Gamma \vdash B}$$

Generalization:

$$\frac{\Gamma \vdash A(x)}{\Gamma \vdash (\forall x) A(x)}, \quad x \text{ not free in } \Gamma$$

- Γ is a set of hypotheses. A is derivable from Γ (written $\Gamma \vdash A$) if some sequence of formulas ends in A , each an axiom, a member of Γ , or obtained from earlier ones by these two rules.
- When $\Gamma = \emptyset$, A is called a theorem, written $\vdash A$.

Worked example: Socrates is mortal

$(\forall x)(P(x) \rightarrow M(x)) \vdash P(\text{Socrates}) \rightarrow M(\text{Socrates})$ by schema 4, $x := \text{Socrates}$. Then
 $P(\text{Socrates}) \wedge [P(\text{Socrates}) \rightarrow M(\text{Socrates})] \vdash M(\text{Socrates})$ by modus ponens.

Inference Rules and Equivalences

Inference rules

Modus Ponens: $P, P \Rightarrow Q \vdash Q$

Modus Tollens: $\neg Q, P \Rightarrow Q \vdash \neg P$

And-Elimination: $P \wedge Q \vdash P$

Or-Introduction: $P \vdash P \vee Q$

Logical equivalences

De Morgan: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

Double negation: $\neg\neg P \equiv P$

Implication elim.: $P \Rightarrow Q \equiv \neg P \vee Q$

Distributivity, commutativity, ...

Conjunctive Normal Form (CNF)

- A sentence in CNF is a conjunction (AND) of clauses
- Each clause is a disjunction (OR) of literals
- **SAT solvers require input in CNF**

Eliminate biconditional: $a \Leftrightarrow b = (a \Rightarrow b) \wedge (b \Rightarrow a)$

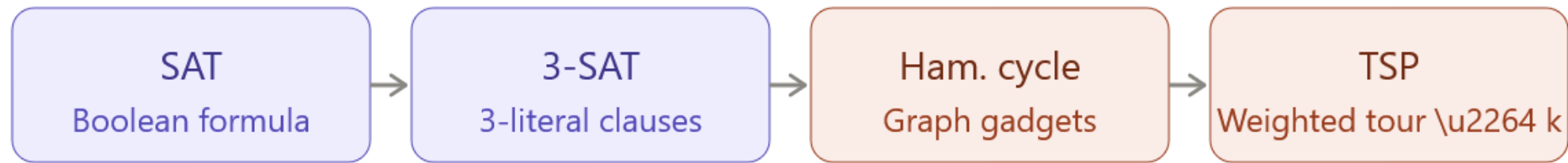
Eliminate implication: $a \Rightarrow b = \neg a \vee b$

Push negation inward (De Morgan's): $\neg(a \wedge b) = \neg a \vee \neg b$

Distribute OR over AND: $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

Connection between SAT (Boolean Satisfiability) and TSP (Traveling Salesman Problem)

Cook\Levin (1971): SAT is the first problem proven NP-complete



Each arrow is a polynomial-time reduction
so all four problems are NP-complete and equally hard

A fast algorithm for any one of these would give a fast algorithm for every problem in NP

SAT: given a propositional formula, does there exist an assignment of T/F to its variables that makes the whole formula true?

TSP: is there a tour visiting all cities with total length $\leq k$?

Why Modern SAT Solvers Scale

~100

variables — naive DPLL's practical ceiling

1000s

clause-learning avoids repeating conflicts

Millions

of variables — modern solvers, real workloads

- Heuristics: smart variable/value ordering, like CSP search
- Clause learning: record conflicts as new clauses, never repeat a mistake
- **Widely used today: hardware verification, dependency resolution, planning**

Logic and SAT in the Real World

Dependency resolvers

apt, pip, npm all solve
a SAT problem to find
a consistent set of
package versions

Chip verification

SAT solvers formally
verify hardware designs
before billions of dollars
go into fabrication

Classical expert systems

MYCIN, DENDRAL: forward/
backward chaining over
hand-written rules — next
lecture's starting point

Resolution: Inference for Clauses

- Propositional resolution: from $(P \vee A)$ and $(\neg P \vee B)$, infer $(A \vee B)$
 - resolve two clauses on a pair of complementary literals
- **FOL resolution adds one step: the complementary literals must first unify**

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From  $(P(t_1, \dots, t_n) \vee A)$  and  $(\neg P(s_1, \dots, s_n) \vee B)$ , if  $\theta = \text{UNIFY}(P(t\dots), P(s\dots))$ ,  
infer  $(A \vee B)\theta$ 
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- **Refutation strategy: negate the goal, convert everything to CNF, resolve**
 - repeatedly until the empty clause $\square$ (a contradiction) is derived
 - $\square$ means the negated goal is unsatisfiable — so the original goal is proved

Can an Omnipotent, Benevolent God Coexist with Sorrow?

Predicates

$E(x)$ x is on Earth

$S(x)$ x is a sorrow

$G(x)$ x is a God

$B(x)$ x is benevolent

$O(x)$ x is omnipotent

$W(x,y)$ x wishes to eliminate y

$P(x,y)$ x has power to eliminate y

$\text{Elim}(x,y)$ x eliminates y

Premises

1. $\exists x (S(x) \wedge E(x))$ — sorrows exist on Earth

2. $\forall x \forall y (G(x) \wedge B(x) \wedge S(y) \wedge E(y) \Rightarrow W(x,y))$ — benevolence

3. $\forall x \forall y (G(x) \wedge O(x) \wedge S(y) \wedge E(y) \Rightarrow P(x,y))$ — omnipotence

4. $\forall x \forall y (W(x,y) \wedge P(x,y) \Rightarrow \text{Elim}(x,y))$ — will+power

5. $\forall x \forall y (\text{Elim}(x,y) \Rightarrow \neg(S(y) \wedge E(y)))$ — elimination

Goal: $\neg \exists x (G(x) \wedge B(x) \wedge O(x))$ — no god is both

Negate the Goal, Convert to CNF

- Negate the goal: $\exists x(G(x) \wedge B(x) \wedge O(x))$ — Skolemize with fresh constant g
- Premise 1 is also existential — Skolemize with fresh constant s

```

C1: G(g)           C2: B(g)           C3: O(g)           (negated goal)
C4: S(s)           C5: E(s)           (premise 1)

C6:  $\neg G(x) \vee \neg B(x) \vee \neg S(y) \vee \neg E(y) \vee W(x, y)$       (benevolence, 2)
C7:  $\neg G(x) \vee \neg O(x) \vee \neg S(y) \vee \neg E(y) \vee P(x, y)$       (omnipotence, 3)
C8:  $\neg W(x, y) \vee \neg P(x, y) \vee \text{Elim}(x, y)$       (will+power, 4)
C9:  $\neg \text{Elim}(x, y) \vee \neg S(y) \vee \neg E(y)$       (elimination, 5)

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The Resolution Refutation

Clause	Resolve	Unifier	Result
C10	C6 with C1, C2	$\{x/g\}$	$\neg S(y) \vee \neg E(y) \vee W(g,y)$
C11	C10 with C4, C5	$\{y/s\}$	$W(g,s)$ — god g wishes to end sorrow s
C12	C7 with C1, C3	$\{x/g\}$	$\neg S(y) \vee \neg E(y) \vee P(g,y)$
C13	C12 with C4, C5	$\{y/s\}$	$P(g,s)$ — god g has power to end sorrow s
C14	C8 with C11, C13	$\{x/g, y/s\}$	$\text{Elim}(g,s)$ — god g eliminates sorrow s
C15	C9 with C14	$\{x/g, y/s\}$	$\neg S(s) \vee \neg E(s)$
C16	C15 with C4	—	$\neg E(s)$
□	C16 with C5	—	contradiction — goal is proved